HARISH G

Embedded Systems and Firmware Engineer

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PROFESSIONAL SUMMARY

Embedded Systems and Firmware Engineer with production experience in bare-metal embedded software development, industrial power electronics, and real-time embedded applications. Proficient in STM32, PIC18, and Nuvoton ARM microcontrollers with hands-on expertise in serial communication protocols and real-time control systems. Delivered industrial-grade control software for servo and static voltage stabilizer systems and power electronics. Reduced manual programming time by 30% through Python automation and directly supported a 40x production scale-up. Recognized with the Best Outgoing Student and Academic Excellence awards. Strong interest in IoT, space technology, and real-time embedded engineering.

TECHNICAL SKILLS

Programming Languages: Embedded C, C, Python, Bare-Metal Programming, RTOS
Microcontrollers: STM32G0, STM32F4, PIC18F46K22, PIC16, Nuvoton M051, Arduino, ESP32
Communication Protocols: UART, I2C, SPI, RS485, CAN, ADC, EEPROM, RTC
Power Electronics: Servo Voltage Stabilizers, Static Voltage Stabilizers, Relay Logic, IGBT Switching
Embedded Software: Firmware Architecture, Interrupt-Driven Design, Fault Protection, Sensor Interfacing, Memory Mapping
Hardware: Circuit Debugging, PCB Analysis, Hardware Integration, Digital Electronics
Tools: Python Automation, Serial Communication Tools, Linux, MPLAB IDE, STM32 CUBE IDE, PROCTEUS, Blender
Soft Skills: Leadership, Resilience**,** Cross-functional Collaboration, Technical Communication

PROFESSIONAL EXPERIENCE

Junior Embedded Developer *Jun 2025 - Dec 2025*
Crisp System India Pvt. Ltd., Coimbatore

- Developed bare-metal embedded software for single-phase and three-phase servo voltage stabilizers, delivering precise voltage regulation across 3 commercial product variants.
- Designed and validated 5 custom control boards through hardware testing, securing a production order of 200 units and achieving a 40x increase in output scale.
- Built multi-board serial communication system for three-phase static stabilizers, enabling reliable data exchange across up to 3 interconnected boards.
- Architected real-time fault logging covering overload detection, earth voltage faults, phase sequence errors, buzzer alerts, and relay cutoff for industrial and domestic operating modes.
- Automated serial number programming using STM32 microcontrollers and Python with protected flash memory mapping and unique ID generation, cutting manual programming time by 30%.
- Implemented relay-based voltage regulation control software with real-time sensing, temperature monitoring, and fault logging for 3-phase stabilizer systems.
- Partnered with hardware, PCB, and production teams to diagnose circuit-level faults and accelerate product deployment timelines.

Research and Development Intern *Sep 2024 - Nov 2024*
Ozotec Automobile Pvt. Ltd., Coimbatore

- Contributed to electric vehicle cluster embedded software development across 3 prototype iterations, completing hands-on system design, electronic prototyping, and hardware troubleshooting.
- Evaluated 5 hardware development tools and streamlined product workflows, improving prototyping efficiency and reducing review cycles by an estimated 20%.

Project Intern *Dec 2024*
CIRPI Education Foundation, Chennai

- Produced educational project documentation and technical content for 2 modules, delivered on schedule across a 1-month engagement.
- Coordinated with 3 team members to meet all project milestones, strengthening project management and stakeholder communication skills.

KEY PROJECTS

1. Three-Phase Servo Stabilizer Controller with LCD Display | *STM32 and PIC18 Microcontrollers | Bare-Metal Production Software*

- Wrote complete bare-metal control software for single and three-phase servo voltage stabilizers, shipped across 3 product variants with real-time voltage and current display and servo motor-based regulation.
- Integrated multi-mode fault protection covering overload, earth voltage, phase sequence, buzzer alerts, and relay cutoff for industrial and domestic environments.

2. Industrial Automatic Relay-Based Voltage Stabilizer | *Nuvoton ARM Microcontroller | Interrupt-Driven Embedded Software*

- Built relay-based tap switching via transformer ratios with real-time voltage, current, and temperature monitoring using analog sensing and non-volatile memory fault logging.

3. Serial Communication System for Static Stabilizers | *STM32 Microcontroller | RS485 Serial Protocol*

- Established scalable multi-board serial communication with automatic unique board ID assignment, supporting three-phase static stabilizer deployments.

4. Automated Serial Number Programming System | *STM32 and Python Automation*

- Constructed a production-line tool for STM32 microcontrollers with protected flash memory mapping, serial ID generation, and automated programming, achieving a 70% reduction in manual effort.

5. AI-Powered Guider for Patients with Alzheimer's Disease | *IoT Solution*

- Delivered an embedded solution with sensor modules and real-time alert systems to assist patients with Alzheimer's Disease in navigation and safety monitoring.

6. Advanced Line-Following Robot | *Arduino Microcontroller | Custom PID Control Algorithm*

- Formulated a custom PID-inspired control algorithm in C and competed in collegiate technical events, achieving top rankings.

7. Smart Blind Stick and Remote Control Car | *Ultrasonic Sensing and RF Communication*

- Smart Blind Stick: Integrated ultrasonic sensors with audio and vibration feedback for real-time obstacle detection to assist visually impaired users.
- Remote Control Car: Assembled a microcontroller-free vehicle using RF encoder and decoder circuits and discrete digital logic, demonstrating strong hardware design fundamentals.

EDUCATION

Bachelor of Engineering - Electronics and Communication Engineering *Nov 2021 - May 2025*
Tamilnadu College of Engineering, Coimbatore
CGPA: 8.1 / 10

Higher Secondary Certificate *2021*
Annai Vailankanni Matric Higher Secondary School
GPA: 90.7%

Secondary School Leaving Certificate *2019*
Government High School
GPA: 90.2%

CERTIFICATIONS

- Embedded System Design - NPTEL, IIT Madras (April 2024)
- Introduction to Programming in C - NPTEL, IIT Madras (April 2024)

ACHIEVEMENTS AND AWARDS

- Best Outgoing Student of the Year - Randeep Memorial Award, Tamilnadu College of Engineering (May 2025)
- Academic Excellence Award - Tamilnadu College of Engineering (May 2025)
- District-Level Chess Championship - 2nd Place
- Technical Competition Certificate - Advanced Line-Following Robot, Collegiate Technical Event